

Edge-Capable Open-Weight LLMs in Modern Greek NLP

A Reproducible Benchmark

John Pavlopoulos Juli Bakagianni

Athens University of Economics and Business / Archimedes, Athena RC
Information Technologies Institute, CERTH

SETN 2026 — Chania, Greece — September 9–11, 2026

Which Greek resources can we actually benchmark?

Resource question

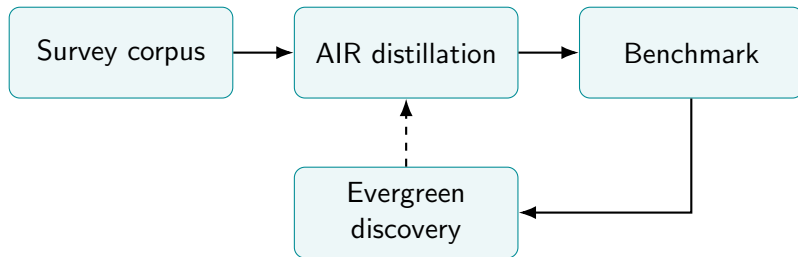
Which Greek datasets are accessible, machine-actionable, and reusable?

Model question

How do six locally served open-weight LLMs perform across Greek NLP tasks?

- ▶ High-quality means native, human-authored text, not synthetic or machine-translated.
- ▶ Resources are fragmented across tasks, domains, and licenses.
- ▶ A shared protocol makes model strengths and weaknesses visible.

From a survey to a reusable evaluation pipeline



- ▶ **Curate:** open-access Greek datasets that permit derivatives, in one recompilable collection.
- ▶ **Classify:** which are usable for benchmarking, which only for pre-training/RAG.
- ▶ **Benchmark:** edge-capable open-weight LLMs, revealing task-specific strengths and weaknesses.

Resource distillation: 94 datasets to 10 retained resources

Survey-based selection

142 studies (2012–2023)

94 datasets analyzed

14 meet initial criteria

10 retained resources

Two distinct uses

7 downstream datasets

Ground truth for task evaluation.

3 pre-training / RAG resources

No gold labels or uncertain automatic annotations.

Exclusions include no-derivatives restrictions, tweet redistribution concerns, a very small corpus, and OCR artifacts.

Seven task families, nine evaluation segments

Family	Dataset / setting	Test size	Primary metric
Intent	University help-desk queries	436	Macro-F1
Legal classification	47 legal volumes	500	Macro-F1
POS tagging	Greek UD treebank	456	Macro-F1
NER	Four entity types	500	Macro-F1
Error correction	Greek Native Corpus	175	GLEU
Translation	Greek → English / Persian / Japanese	3 × 500	chrF
Summarization	Greek court decisions	300	BERTScore-F1

Seven families, nine segments

Translation is evaluated on three language pairs, so it contributes three segments.
Each family keeps the metric that suits it.

Six models under a shared local inference protocol

Open-weight models, 7B–9B

- ▶ Qwen2.5 7B
- ▶ Llama 3.1 8B
- ▶ Llama-Krikri 8B
- ▶ Aya Expanse 8B
- ▶ Gemma2 9B
- ▶ Falcon3 7B

Shared evaluation

- ▶ Ollama; 4-bit Q4_K_M builds.
- ▶ RTX 2080 Ti, 11 GB VRAM.
- ▶ Identical zero-shot prompts.
- ▶ Temperature 0; seed 42 for data.
- ▶ Maximum output: 256 tokens.
- ▶ Summaries: inputs cut to the median test length (13,408 chars).

Model-default Ollama context windows; one workstation, not a survey of edge devices.

Capped tests keep reruns tractable and reproducible

- ▶ Use the first N examples in a fixed test-set order.
- ▶ Retain provider splits; use seed-42 partitions where needed.
- ▶ Preserve all classes in the label-bearing tasks.
- ▶ Label-distribution divergence from full tests: $\text{JSD} \leq 0.015$ bits.

What this establishes

Reproducible subsets with similar label distributions.

It does not establish equivalence to full-test evaluation.

Gemma2 leads the aggregate; no model wins everything

Model	Normalized quality	Segment wins	Avg. latency (s)
Gemma2 9B	77.6	5	1.50
Llama 3.1 8B	61.0	1	1.16
Qwen2.5 7B	49.1	1	1.40
Llama-Krikri 8B	48.5	2	10.40
Aya Expanse 8B	48.0	0	1.14
Falcon3 7B	15.9	0	1.34

Read the aggregate carefully

Nine equally weighted segments: translation contributes three.

Equal weighting of seven families still puts Gemma2 first and Llama second.

Task-specific winners are more informative than one rank

Gemma2: five segment wins

Intent: macro-F1 **0.737**

NER: macro-F1 **0.122**

POS: macro-F1 **0.486**

English MT: chrF **61.781**

Japanese MT: chrF **16.726**

Complementary strengths

Llama 3.1: GEC GLEU **0.796**

Qwen2.5: Persian chrF **18.325**

Krikri: legal macro-F1 **0.156**

Krikri: summary BERTScore **0.533**

Winning a segment does not imply that the task is solved.

Greek adaptation helps on the two legal-domain tasks

Legal classification: macro-F1

Krikri: **0.156**

Gemma2: 0.126

47 labels; imbalanced classes.

Consider richer label descriptions and task-specific adaptation.

These results are consistent with useful specialization; training-data overlap cannot be verified from public documentation.

Summarization: BERTScore-F1

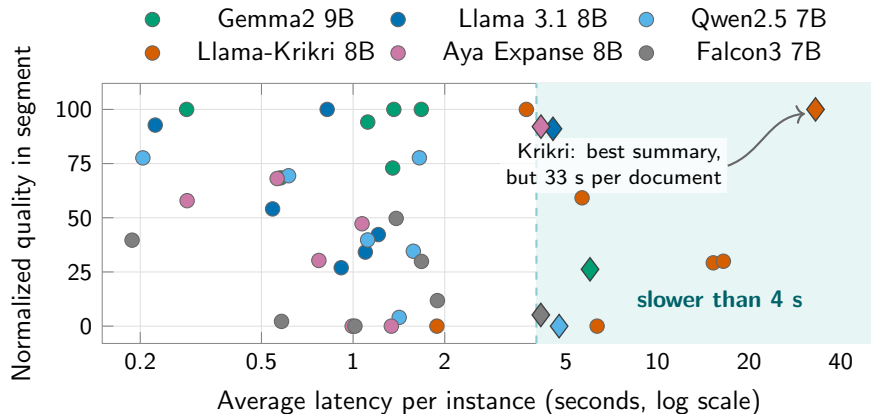
Krikri: **0.533**

Aya: 0.519

Llama 3.1: 0.517

Krikri averages about **33 s** per summary.

Quality and latency must be considered together



Each point is one model on one task family; **diamonds** are summarization. Height ranks models *within* a segment (100 = best there), not absolute difficulty.

Robustness checks support a bounded conclusion

- ▶ **Task weighting:** equal weight per task family keeps Gemma2 first (80.3) and Llama 3.1 second (63.0); the middle three sit within about two normalized points and swap order.
- ▶ **Intent prompt paraphrase:** top two retain their order; mean absolute macro-F1 change is 0.046.
- ▶ **Sensitivity is uneven:** Krikri changes by about +0.154 on that prompt check.

Limits

Capped tests, one prompt check on one task, model-default contexts, and automatic generation metrics. Human assessment remains needed.

Evergreen discovery finds candidates for the next version

118 papers $\longrightarrow$ 55 in scope $\longrightarrow$ 48 dataset indications

18 candidate URLs $\longrightarrow$ **13 reachable**

ACL Anthology and Semantic Scholar, 2024–2026.

A small manual audit (18 annotated papers)

Precision: **1.00** Recall: **0.69**

A proof of concept for discovery, followed by manual curation.

Reachability is not licensing approval. Candidates return to AIR screening; current scores stay fixed.

Takeaways: reusable data, shared tests, task-aware choice

1. **Curate before comparing:** ten retained resources, seven benchmark task families.
2. **Gemma2 leads this protocol;** Llama, Qwen and Krikri have complementary task wins.
3. **Choose by task and cost:** legal classification remains hard; stronger evaluation and continued curation are next.

Code, results, and reproducible artifacts

doi.org/10.5281/zenodo.15314882

github.com/greek-nlp/benchmark

Supported in part by Greece 2.0 / NextGenerationEU (MIS 5154714) and
Horizon Europe ELLIOT (Grant Agreement 101214398).